

Institute of Actuaries of India

Subject SA7-Investment and Finance

May 2023 Examination

INDICATIVE SOLUTION

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

Assets Side:

- mortgages (retail)
- bank loans and overdrafts (retail and wholesale)
- credit card debt (retail)
- interbank deposits (retail and wholesale)
- central bank deposits (retail and wholesale)
- investments in government and corporate bonds (retail and wholesale)

Liabilities Side:

- deposit accounts and term deposits (retail and wholesale)
- interbank borrowing (retail and wholesale)
- secured and unsecured bonds issued (retail and wholesale)
- central bank borrowings (retail and wholesale)

[Max 3]

ii)

Credit risk –risk that loans are not repaid or turned into NPAs (Non-Performing Assets)

Liquidity risk –risk that customers do not act independently but panic and want their money back at the same time

Margin risk –risk that the rate paid for short-term deposits varies while the rate charged on long-term lending remains fixed

Re-pricing risk occurs when assets (loans and investments) mature or re-price (i.e. interest rates are reset) at different times than liabilities (deposits and other borrowings)

[Max 2]

iii)

The distinction between retail and wholesale activities can be summarised as follows:

Retail	Wholesale
High volume	Low volume
Low value transactions	High value transactions
Personal customers and small businesses	Large business customers / other financial institutions
Branch networks	Branch network unnecessary
	High level of competition / lower total costs

Many banking institutions may exist to provide both wholesale and retail services. The major distinguishing factor is really the value of transactions i.e. the amounts deposited and loaned.

Services provided by retail banks for corporate customers include:

- provision of overdraft facilities and loans
- cash management schemes
- electronic banking (via EDI, i.e. electronic data interchange)
- leasing and hire purchase
- export and import financing facilities
- international financial transfers
- financial management advice.

Increasingly, these institutions also offer non-banking services (such as fund management)

Wholesale banks are also known as 'investment', 'merchant' or 'Wall Street' banks. They are more numerous than retail banks, and of smaller average size. Much of their activity involves inter-bank transactions and foreign currency business.

The assets of the wholesale banks are dominated by market loans (wholesale loans at market-related rates of interest, usually to other banking institutions) and advances (lending made predominately to the private sector in the form of overdrafts or loans for fixed periods of time). Repo transactions also feature prominently.

Banking services provided include:

- taking deposits / making loans on a wholesale basis
- acceptance of Bills of Exchange
- management and underwriting of capital issues by companies (i.e. taking care of the issue and sale of bonds on behalf of companies)
- management consulting services (especially with respect to financial aspects)
- advice on mergers and takeovers
- balance sheet advisory
- structured finance
- fund management services (for institutional clients)
- trading in foreign exchange markets
- trading in derivative markets.

[Max 5]

iv) Challenges in managing liquidity risk and in preparing liquidity gap report:

The key issues that are relevant in managing the liquidity risk are:

- Whether the liquidity mismatch (surplus or deficit) for the period under consideration is within tolerable limits as per the ALM policy of the bank?
- If not, what are the strategies to be employed to ensure that the mismatch is brought within tolerance levels set?

Banks could remove liquidity risk by matching the duration of assets and liabilities, but this would also remove their main source of profit from borrowing short and lending long.

Sometimes, even if the liquidity mismatch is within tolerance levels set, the bank may still implement strategies to get the maximum benefit in a given market environment subject to adhering to the limits set at all points of time.

This is the active management of liquidity risk in contrast to passively keeping the liquidity mismatch within limits at all points in time without taking any active position.

There are a number of strategies to managing the mismatch. Typical strategies involve the use of trading portfolio; refinance facilities and lines of credit available from various sources, market borrowing, wholesale deposits, securitization -loan sales etc.

Each one of the strategies has to be analyzed in terms of cost and the intended benefit before a strategy is implemented for managing the mismatch. This is an ongoing process for the ALM department.

Apart from managing liquidity on a day-to-day basis, as per regulator's requirements, all banks are required to produce a Contingency Funding Plan (CFP) to manage stress scenarios

and to withstand a prolonged adverse liquidity crisis. Action plans based on the CFP must be established by the ALM to address the situation arising out of the stress situation.

If customers are confident that the bank is sound they are unlikely to all want their money back at the same time, but if they are worried, they demand their deposits back at the same time and the subsequent bank run will destroy any bank.

To maintain confidence banks should hold (much) higher levels of capital and that too with excess liquidity as 'liquidity buffer'. This is a pool of highly liquid assets that can be sold quickly to realise cash in case of a bank run.

The most recent bank regulation effectively requires banks to hold a reserve equivalent to 30 days stressed outflows. This approach recognises that if a bank cannot be saved within 30 days it is likely that it cannot be saved at all.

Central banks can help in a liquidity crisis by becoming the 'lender of last resort'. In doing so, the central bank must separate liquidity risk from credit risk. If the struggling bank is a viable 'going concern', with sufficient (but illiquid) capital in place to absorb its losses, then the central bank would provide liquidity - but this help should come at a price.

In order to encourage banks to think carefully before taking more risks in pursuit of profits, the central bank should charge a high price for its intervention.

Confidence is also preserved by deposit guarantee schemes, but these create moral hazard as it means banks themselves do not need to be as prudent and expecting to rely less on their own capital in a crisis.

Government intervention can stop a liquidity crisis by essentially nationalising a bank. Once the government owns the bank, depositors stop fearing for the return of their money and the bank run stops.

When a government nationalises a bank, it generally provides share capital (or preference share capital) rather than loans like central banks. Share capital ranks below depositors and unsecured creditors; therefore, the government takes a big risk when it does this.

Liquidity risk is categorized into two types; Trading Liquidity Risk and Funding Liquidity Risk. Trading liquidity risk arises from illiquidity of securities in the trading portfolio of the bank.

As liquidity and return are negatively related, i.e., higher returns can be expected by accepting an illiquid security over a liquid one, strict policy measures to contain the extent of illiquidity in the portfolio need to be set up and monitored on a frequent basis.

The funding liquidity risk arises because of mismatch between the timing of cash flow of the assets from that of the liabilities in the balance sheet

An example of funding liquidity risk let us assume a single asset and a single liability in the balance sheet of a bank. The asset is a three-year loan and the liability is a six-month term deposit. The bank would face funding liquidity risk six-months from today when the six-month term deposit matures for payment since the liability in the form of six-month deposit would require a cash out flow while the asset would not return any cash flow (ignoring possible interest flows from the loan) six-months from today.

This funding mismatch has to be managed by either rolling-over the maturing deposit or acquiring a fresh deposit or sourcing money in any other way suitable for the bank.

Liquidity gap report is useful for measuring short-term liquidity risk, the net gap (B-A) is the difference between inflows and outflows. The concept of liquidity gap is simple to understand but there are few issues:

On assets side:

- Most loans are linked to Prime Lending Rate (PLR) but future dates on which the rates would be reset is unknown. There are no predetermined reset dates for the loans and advances linked to PLR
- Whenever PLR changes, the rate on the loans & advances would change instantly without any time lag whereas for an asset linked to a reference rate, say Treasury bill rate with a 3-month reset, we know very clearly that the rate once set for the asset can change only after 3 months and not before that.
- Borrower has Option to prepay in case of Fixed Rate Term loans: the borrowers may exercise their call option to prepay the loan fully or partially when interest rates go down as they would be in a position to get the advantage of lower rates elsewhere.
- Uncertainty of utilisation of OD (Over Draft) limits: a limit is sanctioned for each borrower and the right to borrow up to the limit is vested with the borrower. Banks pay a price in the form of uncertainty of timing of utilization by the borrowers of the unutilized limits as they may be utilized at any time without prior notice.
- As a result of the uncertainty, banks can neither keep the unutilized portion idle as returns may suffer, nor can deploy it in other assets as the requirement may arise at any time unexpectedly.
- This problem of banks has been exacerbated by the behaviour of a few large corporate houses who maintain an unutilized limit and borrow from the money or capital market as the rate would be cheaper than the lending rate of banks. They return to the unutilized portion when the market turns tight or the market as a source of borrowing dry up.
- The assumptions relating to un-availed portion need to be validated based on behavioural (experience) analysis over a period.
- Cash credit and OD do not have any specific maturity and can result in a cash out flow at any time so it is difficult to place them in a definite bucket
- Repayment pattern need to be guessed based on statistical analysis of behavioural (experience) studies as prescribed by regulator
- In the absence of such analysis, the default approach suggested by regulator for non-maturity deposits maybe used as an alternative
- Major investments are medium to long duration portfolios with low flexibility for altering Illiquidity structure

On liabilities side:

- Deposits – Savings / Current: don't have any maturity date; High volatility of balances in case of current account
- Customer has Option to freely introduce or withdraw money at any time so it is difficult to place them in a definite bucket
- Interest rates applied are unrelated to market interest rates
- Term Deposits: Cumulative / Non-Cumulative / Recurring: majority are in Fixed Rates and few Floating Rate (a recent development)

- Customer Option (put option valuable when interest rates go above the contractual fixed rate) The right but not an obligation to prematurely terminate the term deposit at any time during the currency of the deposit.
- This leads to flight of deposits from a bank to its competitor causing an impact on the liquidity position of the bank.
- The reinvestment risk in case of cumulative deposits
- Some of the maturing term deposits are renewed in the same branch. Such renewals prevent the outflow of cash thereby reducing the net gap.
- While the renewal behaviour is positive, it is difficult to predict perfectly the percentage of maturing deposits renewed period after period as the depositor behaviour may vary from time to time depending on a number of factors.
- If a certain percent of renewal of deposits, say 60% is assumed for the preparation of the gap reports and the actual renewal is only say 40% for a period, this would create an unexpected deficit of liquidity which is serious given the nature of banking business which is heavily dependent upon public confidence.
- Surplus of liquidity would be the result when the actual renewal is higher than the expected renewal of deposits. While surplus is viewed favourably from the point of view of the bank's ability to meet the cash out flows, expected return may suffer as deploying unexpected surplus may diminish expected return.
- Uncertain instalment payments in case of recurring deposits

The customer exercising options either on the liability side or on the asset side need not necessarily have an impact on the existing liquidity position when the liability or asset is re-booked with the same institution at new rates. However, this may affect profit margins.

An experience analysis of the embedded options and their impact on the liquidity position need to be analysed as they display appreciable variations in terms of geographic region, the socio-economic profile of the customer, remaining life of the asset or liability, size of the asset or liability, the extent of change in the market interest rates, etc.

Assumptions relating to off-balance Sheet Items:

- Guarantees provided in favour of customers in various forms like Letters of Credit etc. are off-balance sheet in nature as liabilities are contingent upon future events.
- A financial or performance guarantee is a non-funded commitment until invoked by the party in whose favour the bank issued the guarantee.
- Analysis of the timing and magnitude of crystallization of a non-funded commitment into a funded commitment is extremely important to assess the impact of such crystallization on the liquidity position of the bank.

Finally, the static nature of the Gap Report: The time taken to compile the report determines whether it is useful for decision making or not. If time taken is fairly long, then the first few buckets of information would be useless as the period for which the gaps are calculated would have simply elapsed.

Even if the time delay is considerably reduced, the dynamic nature of the business of banking in which a lot of assets and liabilities are contracted on an ongoing basis would make the figures less relevant in the light of new business, changing behaviour of customers, etc.

Hence, we should consider planned new business in the form of expected assets and liabilities in future and the behavioural pattern of customers to take into account the dynamic nature of the balance sheet.

Unless the dynamic nature of positions and behavioural analysis are incorporated into the analysis of liquidity gap, preparation of a meaningful liquidity report would not be possible

[Max
20]

v) Structural balance sheet ratios are used to assess the liquidity position taking into consideration the structure of the assets and liabilities in the balance sheet.

- Volatile Liabilities (VL) are payable up to one year
- Temporary Assets (TmA) are investments / receivables within one year
- Earning Assets (EA) = Total assets – (Fixed assets + Balances in current accounts with other banks + Other assets excluding leasing + Intangible assets)
- Core deposits (CD)= All deposits (including CASA) above 1 year + net worth
- Total assets (ToA)

$(VL - TmA) / (EA - TmA)$	Measures the extent to which volatile money supports banks' basic earning assets. Since the numerator represents short-term, interest sensitive funds, a high and positive number implies some risk of illiquidity.
CD / ToA	Measures the extent to which assets are funded through stable deposit base.
TmA / ToA	Measures the extent of available liquid assets. A higher ratio could impinge on the asset utilisation of banking system in terms of opportunity cost of holding liquidity.
TmA / VL	Measures the cover of liquid investments relative to volatile liabilities. A ratio of less than 1 indicates the possibility of a liquidity problem.
VL/ToA	Measures the extent to which volatile liabilities fund the balance sheet.

- Call borrowing to total borrowing Extent of dependence on call money market for funding the assets.
- Purchased funds to liquid assets: As purchased funds are volatile to market conditions, the extent to which they are matched by liquid assets.
- Core Deposits to Core Assets: Extent to which retail deposits are used to fund core assets such as Loans and approved securities.
- Liquid assets to deposits: Extent of liquid assets available to meet deposit outflows in an abnormal environment.

[Max 5]

vi) Let X = average price at which the trader shorted 125 futures;
K = Strike Price of option; and
S = Settlement price of ABC share on expiry of the FnOs.
We ignore premiums since net aggregate premiums are zero.

- Here, we will consider only ITM (In-the-money) options and short futures positions to calculate payoff since for ATM (At-the-money) and OTM (Out-of-the-money) options payoff will be zero. Payoff for futures is $125(X-S)$; for ITM Calls is $n(S - K)$; and for ITM Puts is $n(K - S)$
- For payoff at $S=100$ we will consider short futures; $K=95$ calls and $K=105$ puts
 $500,000 = 2000(125(X - 100) - 100(100 - 95) - 110(105 - 100))$
 Solving for X, we get $X = 110.40$

Payoff at +/-5% means at $S=100(1-0.05)=95$ and at $S=100(1+0.05)=105$

- For payoff at $S=95$ we will consider short futures; $K=100$ puts and $K=105$ puts
Payoff = $2000 (125 (110.4 - 95) + 125 (100 - 95) - 110 (105 - 95)) = 2,900,000$
- For payoff at $S=105$ we will consider short futures; $K=95$ calls and $K=100$ calls
Payoff = $2000 (125 (110.4 - 105) - 100 (105 - 95) + 125 (105 - 100)) = 600,000$
- Net derivatives exposure (excluding hedged options) = 125 short futures; 35 long calls; and 35 short puts = net 55 short position. Therefore, we expect more payoffs when share price is low. Thus, payoff at 95 is significantly greater than payoff at 105.

[Max 5]

- vii) In FD, amount deposited is locked without any scope of partial withdrawals when needed. Therefore, if the investor does need the money on an immediate basis, he or she can opt for PMW or breaking of the fixed deposit.

Merit:

- Current PMW feature is easy to understand and apply.
- Outcomes of all possible cases are well known in advance to both the FD holder as well as the bank. i.e., both know what is the pay out if PMW is done at any time $t < n$.

Demerit:

- Under inverted yield curve scenario at the time of opening FD, it is possible to get higher interest rate on PMW than what would be applicable if held till maturity.
- The PMW interest rate used does not reflect true or fair MVA (Market Value Adjustment) that should depend on current FD Term structure of interest rates rather than those at the start of the FD.
- Use of old interest rates for PMW will lead to gain/loss to the bank; i.e. market (interest rate) risk in addition to the liquidity risk.

Change required:

- Let P be the principal deposited at start of FD (of term n). At time $t < n$, PMW occurs then instead of paying $P(1 + X_t - 0.5\%)^t$ with market risk the bank should value FD as (ignoring penalty of 0.5%)
- $$P \frac{(1 + X_n)^n}{(1 + Y_{(n-t)})^{(n-t)}}$$
- This formula calculates amount payable at PMW as the original maturity amount payable at maturity, discounted to current time t , using current FD Term structure of interest rates.
- This formula removes the market risk to the bank and passes on to the customer. To account for the liquidity risk, bank may apply a penalty of say 0.5% or 1.0% of the amount calculated using above formula.

[Max 6]

- viii) **Loan against FD:** Instead of PMW of FD, one can avail loan against FD amount. Bank provides this facility and charges interest 1%-2% above the interest paid on the deposit. Banks allow their customers to avail loan up to 90% of their deposit amount. We need to first compare the costs and benefits of the Loan vs. PMW and select the one that is more beneficial.

Before opening a new FD compare the costs and benefits of opening a sweep-in account or FD laddering as explained below:

Sweep-in facility: This facility allows the lender to credit any sum in excess of some amount stipulated from savings account to a sweep-in account. On opening a sweep-in account, you are likely to earn a higher interest on the amount deposited to your account. This facility offers better corpus and ensures immediate cash requirements are met without having to touch regular investments. Partial or full withdrawals does not attract any penalty whatsoever.

FD Laddering: FD laddering is a process where you apply for various fixed deposits with different maturity periods by taking a lump sum amount and dividing them into smaller investments to open multiple fixed deposit accounts.

For example, if you have a lump sum amount of Rs.5 lakh, you can invest it in five different smaller FDs with maturity periods ranging between one year and five years. In this way, you will not only have long-term fixed deposit accounts, but also short-term accounts, for liquidity.

The laddering process helps to meet your immediate financial requirements without any need for PMWs. Even if you need to PMW money, it will be only to the extent of the amount you will need. For example, if you need an amount of say Rs.2 lakhs, you can prematurely withdraw Rs.1 lakh each from two of your FD account. While you may have to pay a penalty for those two accounts, the remaining ones will continue to fetch you interest on your deposits. You can then choose to reinvest on maturity so that you can increase your cash flow.

[Max 4]

[50 Marks]

Solution 2:

i)

1. **Inflation-Linked Bonds:** These bonds offer protection against inflation as the interest rate is adjusted based on inflation levels. They are a useful tool in managing inflation risk, but their returns may be lower than nominal bonds.
2. **Commodities:** Investing in commodities such as gold, oil, and natural gas can provide a hedge against inflation. However, their returns may be volatile and subject to supply and demand shocks.
3. **Real Estate:** Real estate investments can provide an inflation hedge as rents and property values tend to rise with inflation. However, they may be subject to market fluctuations and are not always easily liquidated.
4. **Infrastructure:** Infrastructure investments such as toll roads and airports can provide a hedge against inflation as they are linked to economic growth and inflation. However, they may require significant capital expenditure and are subject to regulatory and political risks.
5. **Treasury Inflation-Protected Securities (TIPS):** These bonds offer principal and interest rate adjustments to compensate for inflation. They are a useful tool in managing inflation risk, but their returns may be lower than nominal bonds.
6. **Inflation-Indexed Annuities:** These annuities offer a guaranteed income stream that is adjusted based on inflation levels. They can provide a hedge against inflation, but their returns may be lower than other annuities.
7. **Equity Investments:** Investing in companies that are expected to benefit from inflation, such as those in the natural resources sector, can provide a hedge against inflation. However, their returns may be subject to market fluctuations and risks.

8. Floating-Rate Debt: These bonds offer an adjustable interest rate that can provide protection against rising interest rates and inflation. However, their returns may be lower than nominal bonds.
9. Foreign Currency Investments: Investing in currencies that are expected to appreciate with inflation, such as those of emerging market economies, can provide a hedge against inflation. However, they may be subject to exchange rate risk and geopolitical risks.
10. Private Equity: Private equity investments in companies that are expected to benefit from inflation, such as those in the energy and natural resources sectors, can provide a hedge against inflation. However, they are illiquid and require significant capital expenditure.
11. Short-term Debt: Investing in short-term debt instruments such as commercial paper can provide a hedge against inflation. However, their returns may be lower than nominal bonds.
12. Inflation-Linked ETFs: These exchange-traded funds invest in a basket of inflation-linked bonds and can provide a hedge against inflation. However, their returns may be lower than nominal ETFs.
13. Inflation Swaps: These derivative instruments can provide protection against inflation by allowing the transfer of inflation risk between parties. However, they are complex and may be subject to counterparty risk.
14. Inflation Derivatives: These derivative instruments can provide protection against inflation by allowing the transfer of inflation risk between parties. However, they are complex and may be subject to counterparty risk.

[Max 10]

- ii)
1. Incorporating ESG considerations into the investment strategy can help to identify and manage long-term risks associated with inflation.
 2. By investing in sustainable companies with strong ESG practices, the company can potentially achieve better returns in the long run, as these companies tend to be more resilient to economic and market downturns.
 3. ESG investing can help to diversify the company's portfolio and reduce its exposure to certain industries or sectors that may be more vulnerable to inflationary pressures.
 4. Companies with strong ESG practices may be better equipped to manage the impact of inflation on their operations and supply chains, reducing the risk of disruption to their business and ultimately benefiting their investors.
 5. Investing in ESG-focused funds or ETFs can help to increase the overall ESG profile of the company's portfolio and align with its sustainability goals.
 6. One key ESG factor to consider is climate change, which can have significant long-term implications for the economy and financial markets.
 7. Environmental factors such as water scarcity, deforestation, and pollution can also have a significant impact on the economy and should be considered in the investment strategy.
 8. Social factors such as labor practices, human rights, and community engagement can also affect the long-term sustainability of companies and should be considered in the investment process.
 9. Governance factors such as board composition, executive compensation, and shareholder rights can also impact a company's long-term success and should be evaluated in the investment process.
 10. Assessing the carbon footprint and energy efficiency of the companies in the portfolio can help to identify potential risks and opportunities related to energy prices and climate change.

11. Investing in companies with strong ESG practices can help to reduce reputational risk for the life insurance company, as these companies are less likely to be associated with controversies or scandals.
12. Incorporating ESG considerations can help to attract and retain customers who are increasingly concerned about sustainability and social responsibility.
13. By investing in companies that are aligned with its ESG goals, a life insurance company can also help to support positive social and environmental outcomes.
14. Active engagement with portfolio companies on ESG issues can help to promote better corporate practices and reduce risk for the life insurance company.
15. Regular monitoring and reporting on the ESG performance of the portfolio can help to ensure that the investment strategy remains aligned with the company's long-term sustainability goals.

[Max 8]

- iii)
1. A life insurance company can balance the need to manage inflation risk with its investment objectives by developing a comprehensive investment strategy that considers both factors.
 2. The company should establish clear investment goals and a risk management framework to guide its investment decisions.
 3. The investment strategy should take into account the company's risk appetite, investment horizon, and the nature of its liabilities.
 4. The company should consider diversifying its portfolio across different asset classes, such as equities, fixed income, and real assets, to reduce the impact of inflation on its assets.
 5. The company should consider investing in inflation-linked securities, such as TIPS or inflation-linked bonds, to provide a hedge against inflation.
 6. The company should regularly monitor and adjust its investment strategy to reflect changes in the economic and market environment.
 7. The company should consider partnering with investment managers with expertise in managing inflation risk to help design and implement its investment strategy.
 8. The company should evaluate the cost and effectiveness of different hedging strategies, such as interest rate swaps or options, to manage inflation risk.
 9. The company should consider the impact of inflation on its investment returns, as well as on its liabilities, and adjust its asset-liability management strategy accordingly.
 10. The company should regularly assess its liquidity needs to ensure that it has sufficient funds to meet its obligations to policyholders in a timely manner.
 11. The company should consider the impact of inflation on its capital position and ensure that it has adequate capital reserves to withstand inflationary pressures.
 12. The company should consider the impact of inflation on its policyholder base and adjust its product offerings and pricing strategy accordingly.
 13. The company should consider the impact of inflation on its distribution channels, such as its sales force and marketing strategy, and adjust these as needed.

[Max 12]

- iv)
1. Establish a clear investment philosophy and strategy that considers both inflation risk and ESG factors, including the company's risk appetite and investment objectives.
 2. Develop an appropriate asset allocation strategy that considers the risk-return trade-off, the company's liabilities and cash flow requirements, and the impact of inflation and ESG factors.
 3. Identify and analyze asset classes that provide potential inflation hedging benefits, such as inflation-linked bonds, real estate, and commodities.

4. Conduct thorough due diligence on potential investments to ensure they align with the company's ESG goals and are consistent with the investment strategy.
5. Implement a robust risk management framework that includes stress testing, scenario analysis, and risk monitoring tools to identify potential risks and mitigate them effectively.
6. Regularly review and adjust the investment strategy as market conditions and the company's risk profile evolve.
7. Ensure that the investment strategy is consistent with the company's overall financial and business strategy and is approved by senior management and the board of directors.
8. Establish appropriate performance benchmarks and evaluate investment performance regularly against these benchmarks.
9. Monitor the portfolio's liquidity and cash flow requirements to ensure that the company can meet its obligations to policyholders.
10. Monitor the impact of inflation on the company's investment returns, assets, and liabilities regularly.
11. Measure the company's ESG performance and track progress against established targets and industry standards.
12. Conduct periodic external reviews of the investment strategy and performance to ensure it remains relevant and effective.
13. Ensure that the investment team is appropriately trained and resourced to manage the portfolio effectively and monitor inflation risk and ESG factors.
14. Maintain effective communication and collaboration with key stakeholders, including policyholders, regulators, and investors.
15. Regularly report to senior management and the board of directors on the investment strategy, performance, and risk management.

[Max 10]

- v)
1. Inflation can erode the real value of a life insurance company's assets over time, reducing their purchasing power and making it challenging to meet long-term obligations to policyholders.
 2. Increase the company's liability amounts, as policyholders may require higher benefits to maintain their standard of living.
 3. Increased investment volatility, making it difficult to accurately predict future investment returns and meet policyholder obligations.
 4. Reduce the value of fixed income securities, such as bonds, which are a key component of a life insurance company's investment portfolio.
 5. Increase the cost of living, leading to higher claims payouts and potentially reducing the company's profitability.
 6. Lead to higher interest rates, which can reduce the value of existing fixed income securities and increase the company's borrowing costs.
 7. Decline in the value of equity securities, such as stocks, as companies may struggle to maintain profits and dividends in a high inflation environment.
 8. Increased competition for resources, such as labor and materials, which can increase costs for the company and potentially impact its profitability.
 9. Currency fluctuations, which can impact the value of the company's foreign investments and potentially increase the cost of imports.
 10. Changes in government policies and regulations, which can impact the company's investment portfolio and profitability.
 11. Changes in consumer behavior, such as increased saving and reduced spending, which can impact the performance of the company's investment portfolio.

12. Changes in interest rate policies by central banks, which can impact the value of the company's fixed income investments.
13. Changes in global economic conditions, such as recessions or booms, which can impact the performance of the company's investment portfolio.
14. Changes in industry-specific conditions, such as increased competition or changes in technology, which can impact the profitability of the company's investments.
15. Changes in investor sentiment, which can impact the performance of the company's investment portfolio and potentially reduce the company's ability to raise additional capital.

[Max 10]
[50 Marks]
